DATA SCIENCE LABORATORY MANUAL

Module IV ? Statistics and Machine Learning

Topic: Correlation and Regression

OVERVIEW & LEARNING OBJECTIVES

Pearson and Spearman correlation, Ordinary Least Squares simple and multiple linear regression models.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Correlation and Regression in modern Data Science workflows.
- Detailed breakdown: Pearson and Spearman correlation, Ordinary Least Squares simple and multiple linear regression mode
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.